Note 4: The critical values of F corresponding to $\alpha = 0.05$ requires table values at 0.025 and 0.975 which are not provided. Hence α is taken as 0.1 in this example.

Example 3.2

A medical researcher claims that the variance of the heart rates (in beats per minute) of smokers is greater than the variance of heart rates of people who do not smoke. Samples from two groups are selected and the data is given below. Using = 0.05, test whether there is enough evidence to support the claim.

Smokers	Non Smokers
$m = 25$	$n = 18$
$s_1^2 = 36$	$s_2^2 = 10$

Solution:

Step 1 : **Null Hypothesis:** $H_0 : \sigma_1^2 = \sigma_2^2$

That is there is no significant difference between the two population variances.

$H_1 : \sigma_1^2 > \sigma_2^2$

That is, the variance of heart rates of smokers is greater than that of non-smokers.

Step 2 : **Data**

Smokers	Non Smokers
$m = 25$	$n = 18$
$s_1^2 = 36$	$s_2^2 = 10$

Step 3 : **Level of significance** $\alpha = 5\%$

Step 4 : **Test statistic**

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} = \frac{S_1^2}{S_2^2}$$

Step 5 : **Calculation**

$$F_0 = \frac{s_1^2}{s_2^2} = \frac{36}{10} = 3.6$$

Step 6 : **Critical value**

$$f_{(m\text{-}1,n\text{-}1),0.05} = f_{(24,17),0.05} = 2.19$$

Step 7 : **Decision**

Since $F_0 = 3.6 > f_{(24,17),0.05} = 2.19$, the null hypothesis is rejected and we conclude that the variance of heart beats for smokers seems to be considerably higher compared to that of the non-smokers.

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